Muhammad Talha Atif

Backend Engineer

+923064391520 ◇ talhaatif573@gmail.com ◇ Lahore, Pakistan ◇ LinkedIn ◇ GitHub ◇ Portfolio

SUMMARY

Backend Engineer with 1+ year of production experience designing and maintaining RESTful APIs, event-driven microservices, and distributed backend systems using Java (Spring Boot) and Node.js the two primary service environments listed for this role. Hands-on experience with Kafka and RabbitMQ messaging systems, WebSocket-based real-time features, Redis caching strategies, Docker containerization, AWS cloud deployments, and GitHub Actions CI/CD pipelines. Familiar with GoLang concurrency patterns and actively building proficiency. Experienced collaborating with React and TypeScript frontend engineers in Agile/Scrum sprint cycles. Strong background in secure service communication, authentication mechanisms (OAuth 2.0, JWT, RBAC), performance optimization, and structured observability logging across distributed services.

WORK EXPERIENCE

Full Stack Engineer Present
MicroTechX Malaysia (Remote)
Developed and maintained Java (Spring Boot) backend services for an enterprise IAM platform designing RESTful APIs, refactoring legacy monolithic modules into distributed microservices, and implementing event-driven workflows across 8+ services.

- Designed Kafka-based event-driven architecture for audit trails and background provisioning workflows configuring consumer concurrency, retry windows, dead-letter queues, and event validation to ensure fault-tolerant, observable message processing across distributed services.
- Resolved a critical production performance bottleneck by refactoring an O(n²) access-request lookup into an O(n) map-based strategy, implemented gRPC inter-service communication for identity resolution; debugged and fixed a Hibernate multi-bag fetch exception causing intermittent service failures.
- Enforced secure service communication patterns including RBAC, input validation, rate limiting, and structured audit logging, wrote unit tests across tile, cart, and request modules using JUnit, maintained GitHub Actions CI/CD pipelines for automated build, test, and deployment cycles.

Full Stack Engineer Jan '26
Digital Upgraders LLC United Arab Emirates (Remote)
Built and maintained production features for a social networking platform serving photographers, videographers, and creative professionals across web and mobile.

- Optimized frontend and API performance: eliminated a 2-second comment button latency by removing a redundant API call, and reduced notification page network load by consolidating multiple API calls into enriched metadata payloads.
- Built Node.js and Express.js REST APIs for Reported Posts, Reported Comments, and content ranking with Supabase (PostgreSQL) as the primary database, rate limiting on all authentication endpoints, and input validation enforcing secure development practices.
- Resolved 20+ production bugs spanning auth validation, notification lifecycle, cache invalidation, and protected route enforcement, maintained unit test coverage across critical API flows and validated fixes against multi-scenario edge cases before deployment

Open Source Developer Feb '26 — Present
CheckStyle Remote

- Contributed to an Apache-maintained Java static analysis tool by submitting a merged pull request improving documentation for the JavadocCommentsTokenTypes module.
- Added and documented the EQUALS token behavior, improving clarity for developers configuring Javadoc comment rules.
- Merged PR: https://github.com/checkstyle/checkstyle/pull/18807 improving documentation in a widely used Java static analysis tool used across thousands of production codebases.

PROJECTS

Moodly- Experience Booking Platform Link Aug '25 — Present

- Architected a real-time mood detection system using voice tone and facial expression analysis, powered by a RabbitMQ event-driven pipeline to dynamically personalize user experiences, significantly increasing satisfaction and retention.
- Built RESTful backend APIs for booking lifecycle management with TypeScript-strict service layers, input validation, secure authentication, and unit-tested business logic supporting seamless integration with the React Native frontend.
- Implemented voice-command interface enabling hosts to create and manage experiences hands-free, accelerating content onboarding with real-time backend synchronization via event-driven architecture.

Memingo - Language Learning Platform Link Sep '24 — Present

- Optimized leaderboard API performance by implementing **Redis caching**, reducing response time from 30 seconds to approximately 90ms and eliminating user drop-off to enable seamless real-time rankings
- Built a real-time object-detection module that identifies items and instantly delivers vocabulary, cultural context, and learning tips, transforming the app into a highly context-aware language learning experience
- Integrated a **Gemini API-powered chatbot** to provide personalized learning paths and resources, accelerating average learner progression

SKILLS

Programming Languages Javascript, TypeScript, SQL, Java
Backend Spring Boot, Node.js, Nest, Express.js, Mongoose, Fast Api, Flask, Microservices, Next.js, Grpc
Frontend Html, CSS, React JS, TailwindCSS, Shadcn UI, Next.js, CSS3
Databases MySQL, MongoDB, SQ Lite, PostgreSQL, Supabase, MariaDB, Redis
Development Tools Android Studio, VS Code, IntelliJ IDEA, Postman, Eclipse, Pycharm
Version Control Github, Git
AI & Third Party Servcies Swagger, Stripe, Open Ai, Gemini, Hugging Face Models
Cloud Firebase, AWS, AWS S3, AWS RDS, Dokploy
Search Redis, ElasticSearch
Methodology Agile, Scrum, Scrum (Software Development), Scrum Methodology, Scrum Best Practices
AI Tools Claude, Gemini, Github Copilot, Cursor
DevOps CI/CD pipelines, Github Actions, Docker, Kubernetes
Event Driven Kafka, RabbmitMQ
Observability Structured logging, Audit Trails, Error tracking
Testing JUnit, Jest

CERTIFICATIONS

Master Microservices with SpringBoot, Docker, Kubernetes, Udemy — May '25

AWARDS

Runner-up, App Development Hackathon (2025) — May '25
Organized by Punjab University

Participant, Devathon Speed Programming Contest (2024) — Oct '24
Organized by Devsinc

Participant, CALICO Berkeley Speed Coding Competition (2024) — Dec '24
University of California, Berkeley

Ranked, ICPC Asia Topi Online Programming Contest (2024) — Nov '24
Organized by International Collegiate Programming Contests

EDUCATION

Bachelor's degree in Computer Science, University of Management and Technology - UMT — Oct '21 — Aug '25
Lahore, Pakistan